# **Chapter 1:** Discrete Signals and Systems[i].

## **Signal**:

We know that a signal can be a rather abstract notion, such as a flashing light on our bike (turn signal), or a referee's whistle indicating start, halt or end of football match, doorbell ring, etc. Signal can be defined as a **detectable physical quantity or impulse** (as a voltage, current or magnetic field strength) by which **messages or information** can be transmitted.

A signal is a source of information generally a physical quantity which varies with respect to time, space, temperature like any independent variable. In other word, a *signal* is function of independent variables that carry some information.

Eg.$x(t) = 10t + 7t^2$. →Signal of one independent variable (i.e. time)

$s(x,y) = 3x + 2xy + 10y^2$. →Signal of two independent variables (i.e. x and y).

Speech, electrocardiogram (ECG), electroencephalogram (EEG), electromyogram (EMG), etc signals are examples of information-bearing signals that evolve as functions of a single independent variable, namely, time. An example of signal that is function of two independent variables is an image signal.

## **System:**

A system may be defined as a physical device that performs an operation on a signal. More specifically, a system is something that can **manipulate, change, record, or transmit** signals. In Digital Signal Processing, a system may be defined by algorithm. For example, a filter used to reduce the noise and interference corrupting a desired information-bearing signal is called a system. In this case the filter performs some operation/s on the signal, which has the effect of reducing (filtering) the noise and interference from the desired information-bearing signal.

## **Signal Processing vs Signal Analysis:**

When we pass a signal through a system, as in filtering, we say that we have processed the signal. In this modern world we are surrounded by all kinds of signals in various forms. Some of the signals are natural, but most of the signals are manmade. Some signals are necessary (speech), some are pleasant (music), while many are unwanted or unnecessary in a given situation. In an engineering context, signals are carriers of information, both useful and unwanted. Therefore, extracting or enhancing the useful information from a mix of conflicting information is a simplest form of signal processing. More generally, signal processing is an operation designed for extracting, enhancing, storing, and transmitting useful information. The distinction between useful and unwanted information is often subjective as well as objective. Hence signal processing tends to be application dependent.

**Signal analysis** is something we do to design or understand something or a phenomenon. It usually involves mathematical tools as well as devices such as spectrum analyzers, data analysis tools, analytical tools such as FFT, MATLAB, etc. For example, while designing a link, you will examine its spectral spreading and power levels. This is what an engineer will do. But once, he has done with the design, the device itself will just do "signal processing".

Digital Signal Processing refers to methods of filtering and analyzing time-varying signals based on the assumption that the signal amplitudes can be represented by a finite set of integers corresponding to the amplitude of the signal at a finite number of points in time. Digital Signal Processing is distinguished from other areas in computer science by the unique type of data it



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uses: *signals*. In most cases, these signals originate as sensory data from the real world: seismic vibrations, visual images, sound waves, etc. DSP is the mathematics, the algorithms, and the techniques used to manipulate these signals after they have been converted into a digital form. This includes a wide variety of goals, such as: enhancement of visual images, recognition and generation of speech, compression of data for storage and transmission, etc.

Digital signal processing is the study of signals in a digital representation and the processing methods of these signals. DSP includes subfields like: audio signal processing, control engineering, digital image processing and speech processing. RADAR Signal processing and communications signal processing are two other important subfields of DSP.

Since the goal of DSP is usually to measure or filter continuous real-world analog signals, the first step is usually to convert the signal from an analog to a digital form, by using an analog to digital converter. Often, the required output signal is another analog output signal, which requires a digital to analog converter.

The algorithms required for DSP are sometimes performed using specialized computers, which make use of specialized microprocessors called digital signal processors (also abbreviated *DSP*). These process signals in real time and are generally purpose-designed application-specific integrated circuits (ASICs). When flexibility and rapid development are more important than unit costs at high volume, DSP algorithms may also be implemented using field-programmable gate arrays (FPGAs).

**Basic Elements of a Digital Signal Processing System:**

The signals that we encounter in practice are mostly analog signals. These signals, which vary continuously in time and amplitude, are processed using electrical networks containing active and passive circuit elements. This approach is known as analog signal processing (ASP), for example, radio and television receivers.

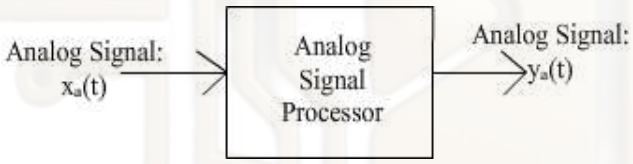



**Figure 1: Analog Signal Processor.**

They can also be processed using digital hardware containing adders, multipliers, and logic elements or using special-purpose microprocessors. However, one needs to convert analog signals into a form suitable for digital hardware. This form of the signal is called a digital signal. It takes of the finite number of values at specific instances in time, and hence it can be represented by binary numbers, or bits. The processing of digital signals is called DSP: in block diagram form it is represented by

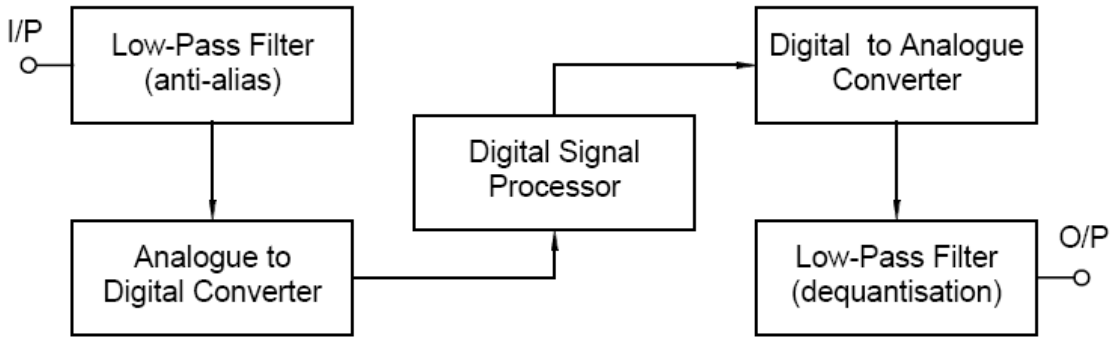



**Figure 2: Basic Elements of Digital Signal Processing.**

The input to and output from the systems are analog in nature. The various block elements are discussed below:



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**Low-Pass Filter (anti-alias):** This is a prefilter or an antialiasing filter, which conditions the analog signal to prevent aliasing.

**Analog to Digital Converter (ADC):** ADC produces a stream of binary numbers from analog signals. It has three sub-elements as sampling, quantization and encoding.

**Digital Signal Processor:** This is the heart of DSP and can represent a general-purpose computer or a special-purpose processor, or digital hardware, and so on. The basic elements of DSP are adder and multiplier.

**Digital to Analog Converter (DAC):** This is the inverse operation to the ADC, which produces a staircase waveform from a sequence of binary numbers, a first step towards producing an analog signal.

**Low-Pass Filter (dequantisation):** This is a postfilter to smooth out staircase waveform into the desired analog signal. It is simply a low pass filter which smooth out (or interpolates) the sequences obtained from DAC.

Performance or these systems is usually limited by the performance (*i.e.* speed, resolution and linearity) of the analog-to-digital converter. However, when using a DSP we should never forget two facts:

- If information was not present in the sampled signal to start with, no amount of digital manipulation will extract it.
- Real signals come with noise.

## Advantages of DSP over ASP:

A major drawback of ASP is its limited scope for performing complicated signal processing applications. This translates into nonflexibility in processing and complexity in system designs. All of these generally lead to expensive products. On the other hand, using DSP approach, it is possible to convert an inexpensive personal computer into a powerful signal processor. Some important advantages of DSP are:

1. Systems using the DSP approach can be developed using software running on a general purpose computer. Therefore DSP is relatively convenient to develop and test, and the software is portable.
2. DSP operations are based solely on additions and multiplications, leading to extremely stable processing capability – for example, stability independent of temperature.
3. Digital signals are easily stored on magnetic or other storable media.
4. The digital signals become transportable and can be processed off-line in a remote laboratory.
5. DSP operations can easily be modified in real time, often by simple programming changes, or by reloading of registers.
6. It is difficult to perform precise mathematical operations on signals in analog form but these same operations can be routinely implemented on a digital computer using software.
7. DSP has lower cost due to VLSI technology, which reduces costs of memories, gates, microprocessors, and so forth.

The principal disadvantage of DSP is the speed of operation of analog to digital converters and DSP, especially at very high frequencies and wide bandwidth require fast-sampling rate A/D converters and fast digital signal processors. Primarily due to the above advantages, DSP is now becoming a first choice in many technologies and applications, such as consumer electronics, communications, wireless telephones, and medical imaging.



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## Applications of Digital Signal Processing:

There are various application areas of digital signal processing (DSP) due to the availability of high resolution spectral analysis. It requires high speed processor to implement the Fast Fourier Transform. Some of these areas can be listed as below:

1. **Speech Processing**

Speech is a one dimensional signal. Digital processing of speech is applied to a wide range of speech problems such as speech spectrum analysis, channel vocoders, etc. DSP is applied to speech coding, speech enhancement, speech analysis and synthesis, speech recognition and speaker recognition.

2. **Image Processing**

Any two-dimensional pattern is called an image. Digital processing of images requires two-dimensional DSP tools such as Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT) algorithms and z-transforms. Processing of electrical signals extracted from images by digital techniques include image formation and recording, image compression, image restoration, image reconstruction and image enhancement.

3. **Radar Signal Processing**

Rader stands for "Radio Detection and Ranging". Improvement in signal processing is possible by digital technology. Development of DSP has led to greater sophistication of radar tracking algorithms. Radar systems consist of transmit-receive antenna, digital processing system and control unit.

4. **Digital Communications**

Application of DSP in digital communication specially telecommunications comprises of digital transmission using pulse code modulation (PCM), digital switching using Time Division Multiplexing (TDM), echo control and digital tape-recorders. DSP in telecommunication systems are found to be cost effective due to availability of medium and large scale digital ICs. These ICs have desirable properties such as small size, low cost, low power, immunity to noise and reliability.

5. **Spectral Analysis**

Frequency-domain analysis is easily and effectively possible in digital signal processing using fast Fourier transform (FFT) algorithms. These algorithms reduce computational complexity and also reduce the computational time.

6. **Sonar Signal Processign**

Sonar stands for "Sound Navigation and Ranging". Sonar is used to determine the range, velocity and direction of targets that are remote from the observer. Sonar uses sound waves at lower frequencies to detect objects under water.

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7. Aviation
8. Astronomy
9. Telecommunication networks
10. Satellite communication
11. Microprocessor systems
12. Industrial noise control.

## Types of Signals:

1. **Continuous time Vs. Discrete-time signal**

As the names suggest, this classification is determined by whether or not the time axis (x-axis) is discrete (countable) or continuous. Continuous-time signals are represented by x(t) where t denotes continuous-time and discrete-time signals or sequences are represented by x[n] where n is an integer denotes discrete-time.



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2. **Continuous value Vs. Discrete-value Signal**

   In discrete-value signals, there are finite values in y-axis. For example, if we look for values between 0 to 1 V, there are infinite numbers of values (Continuous value). But if we fixed the number of values to say 10 (0, 0.1, 0.2, …) or say 5 (0, 0.2, 0.4, 0.6, …) then there are finite numbers of values in y-axis, known as discrete-value signal.

3. **Periodic Vs. Non-Periodic Signal**

   Periodic signals repeat with some period T, while aperiodic or nonperiodic signals do not. We can define a periodic function through the following mathematical expression, where t can be any number and T is a positive constant:

$$\boldsymbol{x(t) = x(t + T)}$$

   *Where T is the fundamental period (the smallest value of T that still allows above equation to be true.*

   For discrete time signal

$$x[n] = x[n + N]$$

   Here, N is the time period which is an integer. That mean the signal repeats after N samples.

4. **Causal Vs. Anticausal Vs. Noncausal**

   Causal signals are signals that are zero for all negative time, while anticausal are signals that are zero for all positive time. Noncausal signals are signals that have nonzero values in both positive and negative time.

5. **Even Vs. Odd**

   An even signal is any signal x(t) such that x(t) = x(-t). Even signals can be easily spotted as they are symmetric around the vertical axis. An odd signal, on the other hand, is a signal that satisfy x(t) = -x(-t) (Also known as Anti-symmetric signal).

   Using the definitions of even and odd signals, we can show that any signal can be written as a combination of an even and odd signal. That is, every signal has an odd-even decomposition.

$$x(t) = x_e(t) + x_o(t)\ (i.e. ever\ part + odd\ part)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}\ \&\ x_o(t) = \frac{x(t) - x(-t)}{2}$$

6. **Deterministic Vs. Random Signal**

   A deterministic signal is a signal in which each value of the signal is fixed and can be determined by a mathematical expression, rule, or table. Because of this the future values of the signal can be calculated from past values with complete confidence. On the other hand, a random signal has a lot of uncertainty about its behavior. The future values of a random signal cannot be accurately predicted and can be usually only be guessed based on the averages of sets of signals.

7. **Energy Vs. Power Signal**

   In electrical systems, a signal may represent a voltage or current. Consider a voltage v(t) developed across a resistor R, producing a current i(t). The *instantaneous power* dissipated in this resistor is defined by

$$p(t) = \frac{v^2(t)}{R}$$

$$or\ equivalently,\ \ p(t) = Ri^2(t)$$

In both cases, the instantaneous power p(t) is proportional to the squared amplitude of the signal. Furthermore, for a resistance R of 1Ω, we see that above equations take on the same mathematical form. Accordingly, in signal analysis it is customary to define power in terms of a 1Ω resistor, so



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that, regardless of whether a given signal x(t) represents a voltage or a current, we may express the instantaneous power of the signal as,

$$p(t) = x^2(t)$$

Based on this convention, we define the *total energy* of the continuous-time signal x(t) as

$$E = \lim_{T\to\infty} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t)dt = \int_{-\infty}^{\infty} x^2(t)dt$$

And its *average power* as

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t)dt$$

In the case of a discrete-time signal x[n], the integrals are replaced by corresponding sums. Thus the total energy of x[n] is defined by

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

And its average power is defined by

$$If\ the\ signal\ is\ aperiodic, \qquad P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

$$\text{If the signal is periodic,} \qquad P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$$

The second expression is the average power in a periodic signal x[n] with fundamental period N. If x[n] is real we can take $|x[n]|^2 = x^2[n]$.

A signal is referred to as an Energy signal, if and only if the total energy of the signal satisfies the condition $0 < E < \infty$.

On the other hand, it is referred to as a Power signal, if and only if the average power of the signal satisfies the condition, $0 < P < \infty$.

The energy and power classifications of signals are mutually exclusive. In particular, an energy signal has zero average power, whereas a power signal has infinite energy. It is also interest to note that periodic signals and random signals are usually viewed as power signals, whereas signals that are both deterministic and non-periodic are energy signals.

**#Find energy or power of the following signals:**

$$\#\ \boldsymbol{x[n] = \left(\frac{1}{2}\right)^n u[n]}$$

The given signal is aperiodic signal and we will find energy as,

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=-\infty}^{\infty} \left|\left(\frac{1}{2}\right)^n u[n]\right|^2 = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^{2n} = \frac{1}{1-\left(\frac{1}{2}\right)^2} = \frac{4}{3}$$

$$\#\ \boldsymbol{x[n] = \cos\left(\frac{\pi n}{4}\right)}$$

The given signal is periodic signal with period N = 8 and we will find power as,

$$P = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n] = \frac{1}{8} \sum_{n=0}^{7} \cos^2\left(\frac{\pi n}{4}\right)$$



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$$= \frac{1}{8}\sum_{n=0}^{7}\left[\frac{1+\cos\left(\frac{\pi n}{2}\right)}{2}\right] = \frac{1}{8}\sum_{n=0}^{7}\frac{1}{2} + \frac{1}{8}\times\frac{1}{2}\sum_{n=0}^{7}\cos\left(\frac{\pi n}{2}\right) = \frac{1}{8}\times\frac{8}{2} + \frac{1}{16}\times 0 = \frac{1}{2}$$

## Continuous-Time Sinusoidal Signals

A simple harmonic oscillation is mathematically described by the following continuous-time sinusoidal signal:

$$x_a(t) = A\cos(\Omega t + \theta), \qquad -\infty < t < \infty$$

This signal is completely characterized by three parameters: *amplitude* A of the sinusoid, *frequency* $\Omega$ in radians per second, and *phase* $\theta$ in radians. Instead of $\Omega$, we often use the frequency F in cycles per second or hertz (Hz), where $\Omega = 2\pi F$.

In terms of F the sinusoid becomes: $x_a(t) = A\cos(2\pi Ft + \theta), -\infty < t < \infty$
The analog sinusoidal signal is characterized by the following properties:

1. For every fixed value of the frequency F, $x_a(t)$ is periodic: $x_a(t + T_p) = x_a(t)$, where $T_p = 1/F$ is the fundamental period of the sinusoidal signal.
2. Continuous-time sinusoidal signals with distinct frequencies are themselves distinct.
3. Increasing the frequency F results in an increase in the rate of oscillation of the signal, in the sense that more periods are included in a given time interval.

The relationships we have described for sinusoidal signals carry over to the class of complex exponential signals

$$x_a(t) = Ae^{j(\Omega t + \theta)}$$

By definition, frequency is an inherently positive physical quantity. This is obvious if we interpret frequency as the number of cycles per unit time in a periodic signal. However, in many cases, only for mathematical convenience, we need to introduce negative frequencies.

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Hence the frequency range for analog sinusoids is $-\infty < F < \infty$.

## Discrete-Time Sinusoidal Signals

A discrete-time sinusoidal signal may be expressed as:

$$x[n] = A\cos(\omega n + \theta), \qquad -\infty < n < \infty$$

Where n is an integer variable, called the sample number, A is the *amplitude* of the sinusoid, $\omega$ is the frequency in radians per second, and $\theta$ is *phase* in radians. Instead of $\omega$, we often use the frequency variable f defined by $\omega = 2\pi f$.



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In terms of f the sinusoid becomes: $x[n] = A\cos(2\pi f + \theta), \quad -\infty < n < \infty$
The frequency f has dimensions of cycles per sample.
In contrast to continuous-time sinusoids, the discrete-time sinusoids are characterized by the following properties:

**1. *A discrete-time sinusoid is periodic only if its frequency f is a rational number:***

By definition, a discrete-time signal x[n] is periodic with period N (>0) iff

$$x[n + N] = x[n] \quad \text{for all } n$$

The smallest value of N for which the above equation is true is called the *fundamental period.*
*Proof:* For a sinusoid with frequency $f_0$ to be periodic, we should have

$$Cos[2\pi f_0(n + N) + \theta] = Cos[2\pi f_0 n + \theta]$$

This relation is true if and only if there exists an integer k such that

$$2\pi f_0 N = 2k\pi \Rightarrow f_0 = k/N$$

Hence, a discrete-time sinusoidal signal is periodic only if its frequency $f_0$ can be expressed as the ratio of two integers (i.e. $f_0$ is rational). To determine the fundamental period N of a periodic sinusoid, we express its frequency $f_0$ as the ratio of two integers and cancel common factors so that k and N are relatively prime. Then the fundamental period of the sinusoid is equal to N.

**2. *Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical.***

*Proof:* Let us consider the sinusoid $\cos[\omega_0 n + \theta]$. It easily follows that

$$\cos[(\omega_0 + 2\pi)n + \theta] = \cos[\omega_0 n + 2\pi n + \theta] = \cos[\omega_0 n + \theta]$$

As a result, all sinusoidal sequences
$x_k[n] = A\cos[\omega_k n + \theta]$ k = 0, 1, 2, 3, ……
where $\omega_k = \omega_0 + 2\pi k \quad -\pi \le \omega_0 \le \pi$

are *indistinguishable (i.e. identical).* On the other hand, the sequences of any two sinusoids with frequencies with frequencies in the range $-\pi \le \omega \le \pi$ or $-\frac{1}{2} \le f \le \frac{1}{2}$ are distinct. Consequently, discrete-time sinusoidal signals with frequencies $|\omega| \le \pi$ or $|f| \le \frac{1}{2}$ are unique. Any sequence resulting from a sinusoid with a frequency $|\omega| > \pi$, or $|f| > \frac{1}{2}$, is identical to a sequence obtained from a sinusoidal signal with frequency $|\omega| < \pi$. Thus we regard frequencies in the range $-\pi \le \omega \le \pi$ or $-\frac{1}{2} \le f \le \frac{1}{2}$ as unique and all frequencies $|\omega| > \pi$, or $|f| > \frac{1}{2}$, as aliases[1].

# $x[n] = \cos(0.6\pi n) \Rightarrow f = 0.3 = 3/10$, rational number hence periodic.

# $x[n] = \cos(0.8n) \Rightarrow f = 0.4/\pi$, irrational number hence aperiodic.

[1]In signal processing and related disciplines, aliasing refers to an effect that causes different signals to become indistinguishable (or aliases of one another) when sampled. It also refers to the distortion or artifact that results when the signal reconstructed from samples is different from the original continuous signal.



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**3. *The highest rate of oscillation in a discrete-time sinusoid is attained when $\omega = \pi$ (or $\omega = -\pi$) or, equivalently, $f = \frac{1}{2}$ (or $f = -\frac{1}{2}$).***

To illustrate this property, let us investigate the characteristics of the sinusoidal signal sequence

$$x[n] = \cos\omega_0 n$$

When the frequency varies from 0 to π. To simplify the argument, we take values of $\omega_0 = 0, \pi/8, \pi/4, \pi/2, \pi$ corresponding to $f = 0, {}^1/_{16}, {}^1/_8, ¼, ½$; which results in periodic sequences having periods $N = \infty, 16, 8, 4, 2$. We note that the period of the sinusoid decreases ( or rate of oscillation increases) as the frequency increases.

To see what happens for $\pi \le \omega_0 \le 2\pi$, we consider the sinusoids with frequencies $\omega_1 = \omega_0$ and $\omega_2 = 2\pi - \omega_0$. Note that as $\omega_1$ varies from π to 2π, $\omega_2$ varies from π to 0. It can be easily seen that

$x_1[n] = A\cos\omega_1 n = A\cos\omega_0 n$
$x_2[n] = A\cos\omega_2 n = A\cos(2\pi - \omega_0)n = A\cos(-\omega_0 n) = x_1[n]$

Hence $\omega_2$ is an alias to $\omega_1$. If we had used a sine function instead of a cosine function, the result would basically be the same, except for a $180^0$ phase difference between the sinusoids $x_1[n]$ and $x_2[n]$. In any case, as we increase the relative frequency $\omega_0$ of a discrete-time sinusoid form π to 2π, its rate of oscillation decreases. For $\omega_0 = 2\pi$ the result is a constant signal, as in the case of $\omega_0 = 0$. Obviously, for $\omega_0 = \pi$ (or $f = ½$) we have the highest rate of oscillation.

Since discrete-time sinusoidal signals with frequencies that are separated by an integer multiple of 2π are identical, it follows that the frequencies in any interval $\omega_1 \le \omega \le \omega_1 + 2\pi$ constitute all the existing discrete-time sinusoids or complex exponentials. Hence the frequency range for discrete-time sinusoids is finite with duration 2π. Usually, we choose the range $0 \le \omega \le 2\pi$ or $-\pi \le \omega \le \pi$ ($0 \le f \le 1$, $-½ \le f \le ½$), which we call the *fundamental range.*

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**Harmonically Related Complex Exponentials**

These are sets of periodic complex exponentials with fundamental frequencies that are multiple of a single positive frequency.

$$s_k(t) = e^{jk\Omega_0 t} = e^{jk2\pi F_0 t} k = 0, \pm 1, \pm 2, - - -$$

$$FundamentalPeriod = \frac{1}{kF_0} = \frac{T_p}{k}$$

Similarly for discrete-time, $s_k[n] = e^{j2\pi k f_0 n}$



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**Analog to Digital Convertor (ADC):**

1. **Sampling**: Continuous-time signal to discrete-time signal

   x(t) ===> x(nT) ≡ x[n], T is the sampling interval.

2. **Quantization**: Discrete-time continuous value signal to discrete-time discrete-value signal.

   x[n] ===> $x_q[n]$. Quantization error, $q_e = x[n] - x_q[n]$.

3. **Coding**: In the coding process, each discrete value $x_q[n]$ is represented by a $b$ – bit binary sequence.

**Sampling of Analog Signal:**

We limit our discussion to periodic or uniform sampling.

$$x(t) = A\cos(2\pi Ft + \theta)$$

$$x[n] \cong x(nT), \qquad -\infty < n < \infty$$

$$t = nT = \frac{n}{F_s}$$

Sampling periodically at a rate $F_s = 1/T$ samples/sec

$$x(nT) \cong x[n] = A\cos(2\pi FnT + \theta) = A\cos\left(2\pi\left(\frac{F}{F_s}\right)n + \theta\right)$$

If we compare with discrete-time sinusoid $x[n] = A\cos(2\pi fn + \theta)$ we get,

$$f = \frac{F}{F_s}, \quad or, \omega = \Omega T$$

Is called relative or normalized frequency. We can use f to determine the frequency F in hertz only if the sampling frequency $F_s$ is known.

Recall,

$$-\infty < F < \infty \& -\frac{1}{2} \le f \le \frac{1}{2} cycles/samples$$



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$$-\infty < \Omega < \infty \& -\pi \leq \omega \leq \pi radians/samples$$

We find that the frequency of the continuous-time sinusoid when sampled at a rate $F_s = 1/T$ must fall in the range

$$-\frac{1}{2T} = -\frac{F_s}{2} \leq F \leq \frac{F_s}{2} = \frac{1}{2T}$$

Mapping of infinite range to finite frequency range of variable f. since highest frequency in a discrete-time signal is $\omega = \pi$ or $f = ½$ with sampling rate $F_s$, the corresponding highest value of F is, $F_{max} = \frac{F_s}{2} = \frac{1}{2T}$

## Some terms related to sampling of analog signals:

**Sampling Frequency, $F_s$**: It is the number of samples per second while converting continuous-time signal to discrete-time signal.

**Nyquist Criteria, $F_s \geq 2F_{max}$**: It is the criteria that has to be fulfilled for the reconstruction of signal from discrete-time signal. $F_s$ is the sampling frequency and $F_{max}$ is the maximum frequency contained in continuous-time signal.

**Nyquist Rate, $F_N = 2F_{max}$**: It is the minimum sampling frequency required for the proper reconstruction of the signal.

**Folding Frequency (or Nyquist Frequency), $F_s/2$**: The highest frequency that can be reconstructed or measured using discretely sampled data. It is the half of sampling frequency.

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**# Consider the two analog sinusoidal signals**

$$x_1(t) = \cos(2\pi(10)t) \& x_2(t) = \cos(2\pi(50)t)$$

Sample the two signals at a rate $F_s = 40$ Hz and find the discrete-time signals obtained.

[Hint: The two sinusoidal signals are identical & consequently indistinguishable. We say that the frequency $F_2 = 50$ Hz is an alias of the frequency $F_1 = 10$ Hz at the sampling rate of 40 Hz]

**# Consider the analog signal**

$$x(t) = 3cos100\pi t$$



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a. Determine the minimum sampling rate required to avoid aliasing.
b. Suppose that the signal is sampled at $F_s$ = 200 Hz, What is the discrete-time signal obtained after sampling?
c. Suppose that the signal is sampled at the rate Fs = 75 Hz, what is the discrete-time signal obtained after sampling?
d. What is the frequency 0 < F < Fs/2 of a sinusoid that yields samples identical to those obtained in part(c) ?

**# Consider the analog signal**

$$x(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12000\pi t$$

a. What is the Nyquist rate for this signal?
b. Assume now that we sample this signal using a sampling rate Fs = 5000 samples/s. What is the discrete-time signal obtained after sampling?

What is the analog signal y(t) we can reconstruct from the samples if we use ideal interpolation?

**# A digital communication link carries binary-coded words representing samples of an input signal,**

$$x(t) = 3cos600\pi t + 2cos1800\pi t$$

The link is operated at 10,000 bits/s and each input sample is quantized into 1024 different voltage levels.

i. What is the sampling frequency & folding frequency?
ii. What is the Nyquist rate for the signal x(t) ?
iii. What are the frequencies in the resulting discrete time signal x[n] ?
iv. What is the resolution Δ ?

Solution: ➔

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As the link is operated at 10,000 bits/s & each input sample is quantized into 1024 different voltage levels the each sampled value is represented by $\log_2 1024$ = 10 bits/sample

i. Then maximum sampling frequency, $F_s = \frac{10{,}000\ bits/sec}{10\ bits/sample} = 1000\ samples/sec$

Folding frequency (is the maximum frequency that can be represented uniquely by sampled signal), $\frac{F_s}{2} = 500\ samples/sec.$

ii. $x(t) = 3cos600\pi t + 2cos1800\pi t$

Here, $F_1$ = 300 Hz and $F_2$ = 900 Hz. Thus $F_{max}$ = 900 Hz.

The Nyquist rate, $F_N = 2F_{max}$ = 1800 Hz.

iii. For $F_s$ = 1000 Hz,



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$$x[n] \cong x(nT) = x\left(\frac{n}{F_s}\right) = 3cos2\pi\left(\frac{300}{1000}\right)n + 2cos2\pi\left(\frac{900}{1000}\right)n$$

$$= 3cos2\pi\left(\frac{3}{10}\right)n + 2cos2\pi\left(\frac{9}{10}\right)n$$

$$= 3cos2\pi\left(\frac{3}{10}\right)n + 2cos2\pi\left(1 - \frac{1}{10}\right)n$$

$$= 3cos2\pi\left(\frac{3}{10}\right)n + 2cos2\pi\left(\frac{1}{10}\right)n$$

$$\therefore f_1 = \frac{3}{10} \& f_2 = \frac{1}{10}$$

$$Here\ both\ frequencies\ f_1 and\ f_2\ lies\ in\ the\ interval\ -\frac{1}{2} \leq f \leq \frac{1}{2}$$

iv. ADC resolution = 10 bits

Voltage resolution, $\Delta = \frac{x_{max} - x_{min}}{L-1} = \frac{5-(-5)}{1024-1} = 9.76\ mV.$

## Discrete-time signals:

Digital signals are discrete in both time (the independent variable) and amplitude (the dependent variable). Signals that are discrete in time but continuous in amplitude are referred to as discrete-time signals. Discrete-time signals are data sequences. A sequence of data is denoted by {x[n]} or simply x[n] when the meaning is clear. The elements of the sequence are called samples. The index n associated with each sample is an integer.

A discrete-time (D.T.) signals x[n] is a function of an independent variable that is an integer. It is important to note that a D.T. signal is not defined at instants between two successive samples. Also, it is incorrect to think that x[n] is equal to zero if n is not an integer. Simply, the signal x[n] is not defined for non-integer values of n.

**Representation of D.T. Signals:**

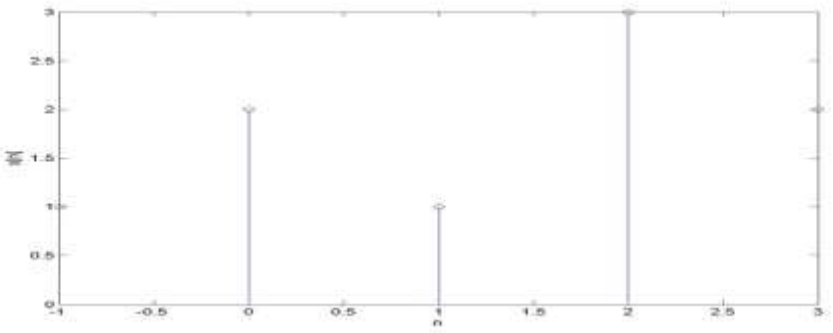
1) Tabular representation

| n | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|
| x[n] | 1 | 2 | 1 | 3 | 2 |

2) Graphical representation



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3) Functional representation

$$x[n] = \begin{cases} 1 \ for\ n = -1, 1 \\ 2\ for\ n = 0, 3 \\ 3 \ for\ n = 2 \\ 0\ elsewhere \end{cases}$$

4) Sequence representation

x[n] = {1, 2, 1, 3, 2} where *" ↑ sign represents the position of n =0. The arrow is often omitted if it is clear from the context which sample is x[0].*

Sample values can either be real or complex. The terms "discrete-time signals" and "sequences" are used interchangeably.

## Some Elementary D.T. signals:

Elementary discrete-time signals can be defined as those signals that can be used as building blocks to build more complex signals and are normally used for the study of signals and systems. Conversely, we can decompose complex signals into elementary signals and analyze the behaviors of signals and systems. Followings are the discrete-time elementary signals we study in this course:

1) The **unit sample sequence** or unit impulse or Kronecker delta is defined as δ[n] and is defined as

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$$\delta[n] = \begin{cases} 1, for\ n = 0 \\ 0, for\ n \neq 0 \end{cases}$$

*Note: The analog signal δ(t) or Dirac delta is defined to be zero everywhere except at t=0 and has unit area.*

2) The **unit step signal** is denoted as u[n] and is defined as

$$u[n] = \begin{cases} 1, for\ n \geq 0 \\ 0, for\ n < 0 \end{cases}$$

**# Prove that**

$$i. u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

$$ii. \delta[n] = u[n] - u[n-1]$$



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3) The **unit ramp signal** is denoted as $u_r[n]$ and is defined as

$$u_r[n] = \begin{cases} n, for\ n \geq 0 \\ 0, for\ n < 0 \end{cases}$$

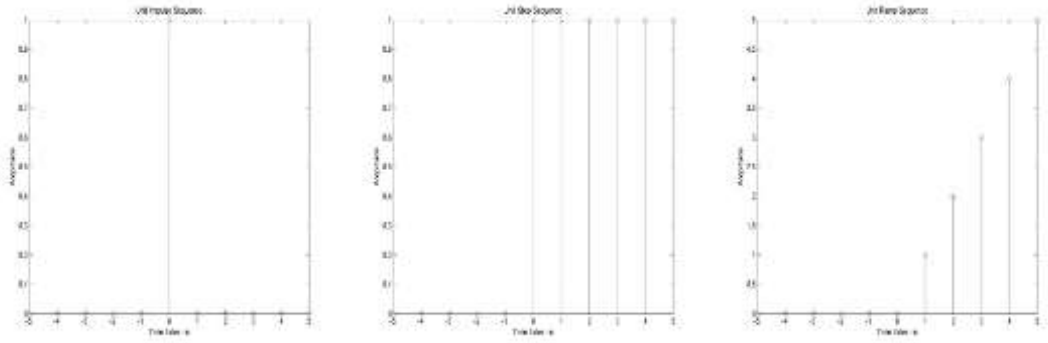


*Fig: (a) Unit impulse (b) Unit Step (c) Unit Ramp sequences.*

**4) Sinusoidal Signal**

$x[n] = A\cos(\omega n + \theta)$, $-\infty < n < \infty$

Where n is an integer variable, called the sample number, A is the *amplitude* of the sinusoid, ω is the frequency in radians per second, and θ is *phase* in radians.

**5) Exponential Signal**

The exponential signal is a sequence of the form

$x[n] = a^n$ for all n.

If the parameter a is real, then x[n] is a real signal. When the parameter a is complex valued,

$a = re^{i\theta}$ (r & θ are now parameters)

Hence, $x[n] = r^n e^{i\theta n} = r^n(\cos\theta n + j\sin\theta n)$

The real part is $x_R[n] = r^n\cos\theta n$ and the imaginary part is $x_I[n] = r^n\sin\theta n$

Alternatively, the complex signal x[n] can be represented by the amplitude function

$|x[n]| = A(n) = r^n$

$/_x[n] = \varphi(n) = \theta n$

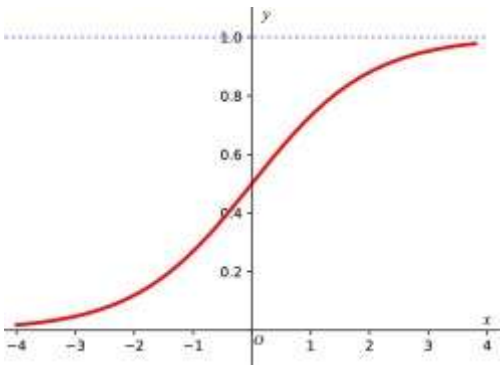
**6) Sigmoid Function**

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The sigmoid function is one of the most commonly used activation functions. It is denoted by σ(x) and given as

$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x} = 1 - \sigma(-x)$$

This function transforms a continuous real number into a range of (0, 1). It is commonly used in neural networks as an activation function, where small input values result in outputs close to 0 and large input values result in outputs close to 1. It is also known as logistic function.








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## Classification of Discrete-time signals:

**1) Energy signals & Power signals:**

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

If E is finite (i.e. 0 < E <∞), then x[n] is *energy signal.*

Many signals that possess infinite energy, have a finite average power. The average power of a periodic sequence with a period of N samples is defined as

$$P = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2$$

And for non-periodic sequences, it is defined in terms of the following limit if it exists:

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

A signal with finite average power is called a power signal.

**# Find the average power of the unit step sequence u[n].**

The unit step sequence is non-periodic, therefore the average power is

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} u^2[n]$$
$$= \lim_{N\to\infty} \left(\frac{N+1}{2N+1}\right) = \frac{1}{2}$$

Therefore, the unit step sequence is a power signal. Note that its energy is infinite and so it is not an energy signal.

**2) Periodic signals & Aperiodic signals:**

The sinusoidal signal of the form
$x[n] = A\sin 2\pi f_0 n$ is periodic when $f_0$ is a rational number, that is if $f_0$ can be expressed as $f_0 = k/N$ ; where k and N are integers and N is the fundamental period.

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**3) Symmetric(even) and antisymmetric(odd) signals:**

Even → x[-n] = x[n] $\quad x_e[n] = ½ \{x[n] + x[-n]\}$

Odd → x[-n] = -x[n] $\quad x_o[n] = ½ \{x[n] - x[-n]\}$

## Transformation of the independent variable (time)

**1) Shifting:**

A signal x[n] may be shifted in time by replacing the independent variable n by n-k, where k is an integer. If k is positive, delay of the signal by k units of time and if k is negative, advance of the signal by k units in time.

**2) Folding or a reflection of the signal about the time origin n=0:**



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A signal x[n] may be folded by replacing the independent variable n by –n.

# Prove that the operations of folding and time shifting a signal are not commutative.

$$TD_k[x[n]] = x[n-k] \quad k>0$$

$$FD[x[n]] = x[-n]$$

Now, $TD_k[FD\{x[n]\}] = TD_k[x[-n]] = x[-n+k]$

Whereas, $FD[TD_k\{x[n]\}] = FD\{x[n-k]\} = x[-n-k]$

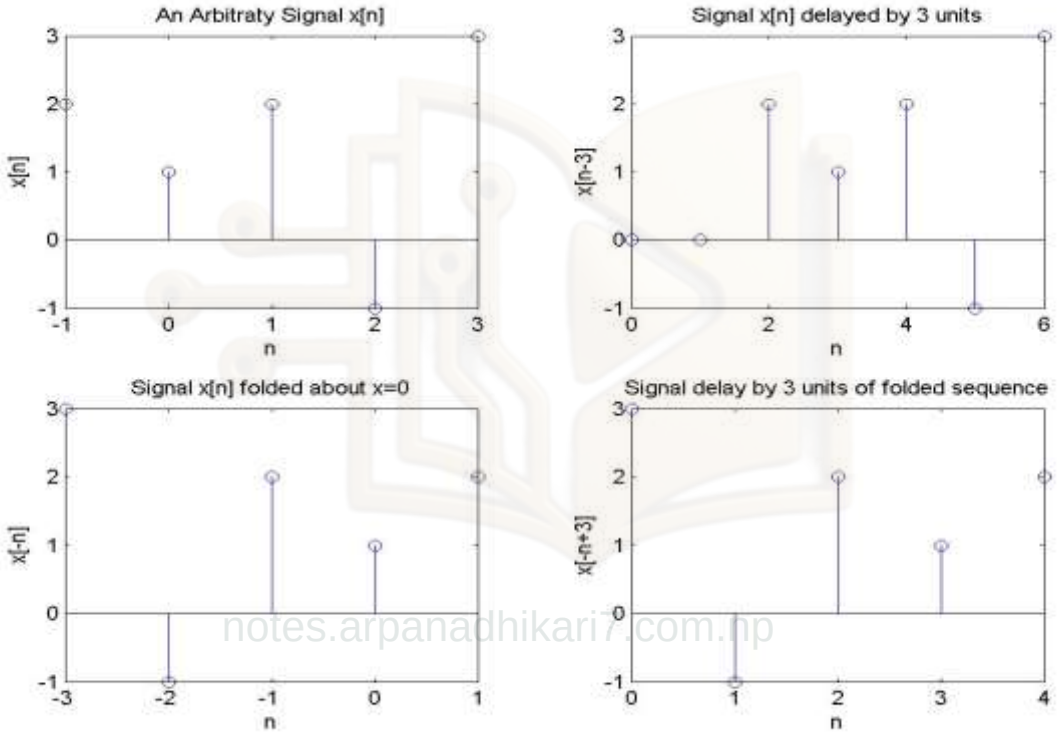
Note: Shifting of folded sequence x[-n] ---> x[-n+k]

If k positive, delay; if k negative, advance.

**3) Time scaling or down-sampling:**

-Replacing n by µn, where µ is an integer.

If the signal x[n] was originally obtained by sampling an analog signal $x_a(t)$, the $x[n] = x_a(nT)$, where T is the sampling interval. Now, $y[n] = x[2n] = x_a(2Tn)$. Hence the time-scaling operation is equivalent to changing the sampling rate from 1/T to 1/2T i.e. decreasing the rate by a factor of 2. This is a downsampling operation. Upsampling is not possible as we cannot obtain y[n] = x[n/2] from the signal x[n].




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Note: Amplitude modification includes addition, multiplication, and scaling of D.T. signals.

## Discrete-time systems:

In many applications of digital signal processing we wish to design a device or an algorithm that performs some prescribed operation on a discrete-time signal. Such a device or algorithm is called a discrete-time system.

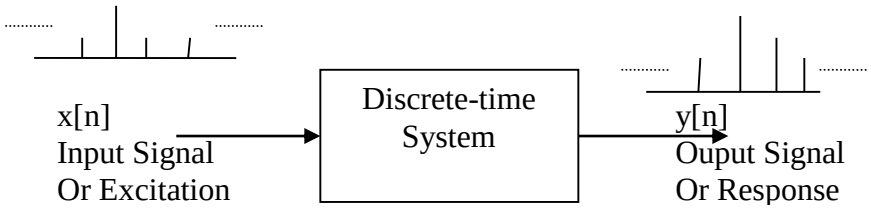
A discrete-time system is a device or algorithm that operates on a discrete-time signal, called the input or excitation, according to some well-defined rule, to produce another discrete-time signal called the output or response of the system.

$$y[n] = T\{x[n]\}$$



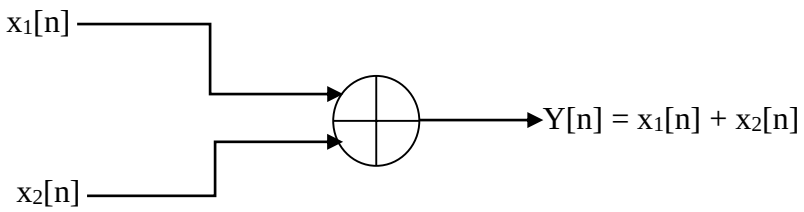
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We say that the input signal x[n] is transformed by the system into a signal y[n].

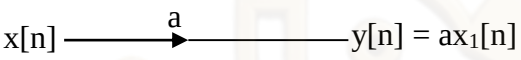



**Block Diagram Representation of DT Systems:**
(Basic building blocks that can be interconnected to form complex systems)

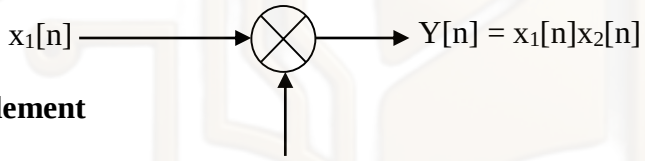
- **An Adder:**



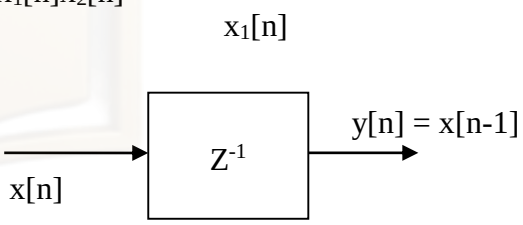

- **A Constant Multiplier**




- **A Signal Multiplier**

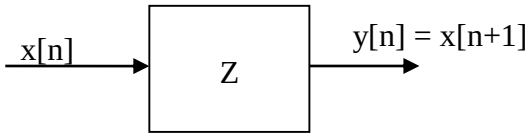



- **A Unit delay element**




- **A Unit advance element**

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#Use basic building blocks to represent following DT system described by the input-output relation:
$y[n] = 0.25y[n-1] + 0.5x[n] + 0.5x[n-1]$

## Classification of Discrete-time systems:

### 1) Static (memoryless) vs dynamic (to have memory) system: -

If the o/p of D.T. system at any instant n depends at most on the input sample at the same time, but not on past or future samples of the input then it is known as static system.
In any other case, the system is said to be dynamic.
$y[n] = x[n-N]$ $N \geq 0$
The system is said to have memory of duration N,



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If N = 0, the system is static
$0<N<\infty$, the system is said to have finite memory.
$N = \infty$, the system is said to have infinite memory.

**2) Time-invariant Vs. time-variant system: -**

A system is called time-invariant if its input-output characteristics do not change with time.
Theorem:
A relaxed system T is time invariant or shift invariant if and only if $x[n] \rightarrow y[n]$
Implies $x[n-k] \rightarrow y[n-k]$
For every input signal x[n] and every time shift k.
For this we compute
$y[n, k] = T\{x[n-k]\}$ → response of delayed input and
$y[n-k]$ → delayed response
and check whether $y[n, k] = y[n-k]$ for all possible values of k.
if true → time invariant
& if $y[n, k] \neq y[n-k]$, even for one value of k, the system is time variant.

**3) Linear Vs. nonlinear system: -**

A linear system is one that satisfies the superposition theorem which states that, "A system is linear if and only if
$T\{a_1x_1[n] + a_2x_2[n]\} = a_1T\{x_1[n]\} + a_2T\{x_2[n]\}$
For any arbitrary i/p sequences $x_1[n]$ and $x_2[n]$, and any aribitrary constants $a_1$& $a_2$.

**4) Causal Vs. noncausal system: -**

Theorem: "A system is said to be causal if the o/p of the system at any time n {i.e. y[n]} depends only on present and past inputs but doesnot depend on future inputs".
If a system doesnot satisfy this definition, it is called noncausal.

**5) Stable Vs. unstable system: -**

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An arbitrary relaxed system is said to be bounded input bounded output (BIBO) stable if and only if every bounded input produces a bounded output.
Mathematically,
If, $|x[n]| \leq M_x < \infty$
Then there must be $|y[n]| \leq M_y < \infty$ for all n, for the system to be stable. Where, $M_x$ and $M_y$ are finite numbers.

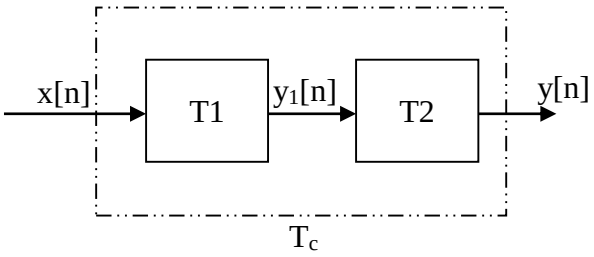
## Interconnection of Discrete-time systems:

Discrete time systems can be interconnected to form larger systems. There are two basics ways in which systems can be interconnected:

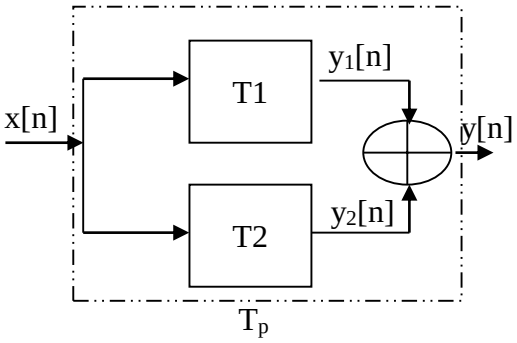
1. Cascade (Series)
2. Parallel



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(a) Cascade Interconnection




(b) Parallel Interconnection

In the cascade interconnection the output of the first systemis

$$y_1[n] = T_1\{x[n]\}$$

and the output of the second system is

$$y[n] = T_2\{y_1[n]\} = T_2[T_1\{x[n]\}]$$

We observe that system $T_1$ and $T_2$ can be combined or consolidated into a single overall system

$$T_c \equiv T_2T_1$$

We can express the output of the combined system as,

$$y[n] = T_c\{x[n]\}$$

In general, for arbitrary systems,

$$T_2T_1 \neq T_1T_2$$

However, if the systems $T_1$ and $T_2$ are linear and time-invariant, then (a) $T_c$ is time invariant and (b) $T_2T_1 = T_1T_2$

Proof of (a)

Suppose that $T_1$ and $T_2$ are time-invariant, then

$x[n-k] \text{ — } (T_1) \rightarrow y_1[n-k]$

$y_1[n-k] \text{ —} (T_2) \rightarrow y[n-k]$

Thus, $x[n-k] \text{—} (T_c) \rightarrow y[n-k]$

And therefore, $T_c$ is time invariant.

In the parallel interconnection, the output of the system $T_1$ is $y_1[n]$ and the output of the system $T_2$ is $y_2[n]$. Hence,

$y_3[n] = y_1[n] + y_2[n] = T_1\{x[n]\} + T_2\{x[n]\} = (T_1 + T_2)\{x[n]\} = T_p\{x[n]\}$.

Where, $T_p = T_1 + T_2$

In general, we can use parallel and cascade interconnection of systems to construct larger, more complex systems. Conversely, we can take a larger system and break it down into smaller subsystems for purposes of analysis and implementation.

## Analysis of Discrete-time linear time-invariant(LTI) systems:

There are two basic methods for analyzing the behavior or response of a linear system to a given input signal.

- **Convolution method:**

  This method for analyzing the behavior of a linear system to a given input signal is first to decompose or resolve the input signal into a sum of elementary signals. Then, using the linearity property of the system, the responses of the system to the elementary signals are added to obtain the total response of the system to the given input signal.

- **Linear constant coefficient difference (LCCD) equation method:**



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This method is based on the direct solution of the input-output equation for the system, and has the form
y[n] = F{y[n-1], y[n-1], ……, y[n-N], x[n], x[n-1], x[n-2], ……, x[n-M] }
Specifically, for an LTI system, the general form of the input-output relationshiop is,

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

Where $\{a_k\}$ and $\{b_k\}$ are constant parameters that specify the system and are independent of x[n] and y[n].

## Resolution of a Discrete-time signal into impulses:

Suppose we have an arbitrary signal x[n] that we wish to resolve into a sum of unit sample sequences. We select the elementary signals $x_k[n]$ to be

$$x_k[n] = \delta[n-k]$$

where k represents the delay of the unit sample sequence.
Now suppose that we multiply the two sequences x[n] and δ[n-k], since δ[n-k] is zero everywhere except at n = k, where its value is unity, the result of this multiplication is another sequence that is zero everywhere except at n = k, where its value is x[k]. Thus
x[n]δ[n-k] = x[k]δ[n-k]
If we were to repeat the multiplication of x[n] with δ[n-m] where m is another delay (m≠k), the result will be a sequence that is zero everywhere except at n = m, where its value is x[m]. Hence,
x[n]δ[n-m] = x[m]δ[n-m]
Consequently, if we repeat this multiplication over all possible delays $-\infty < k < \infty$, and sum all the product sequences, the result will be a sequence equal to the sequence x[n], that is,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

The right hand side of the equation gives the resolution of or decomposition of any arbitrary signal x[n] into a weighted (scale) sum of shifted unit samples sequences.
Response of LTI systems to arbitrary inputs: The Convolution Sum
We know,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

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First, we denote the response y[n ,k] of the system to the input unit sample sequence at n = k by the special symbol h[n, k]; $-\infty < k < \infty$, i.e.
y[n, k] = h[n, k] = T{δ[n-k]}
we note that n is the time index and k is a parameter showing the location of the input impulse. If the impulse at the input is scaled by an amount $C_k = x[k]$, the response of the system is the corresponding scaled output, that is
$C_k h[n, k] = x[k]h[n, k]$
The response to any arbitrary input expressed as a sum of weighted impulses, as given above is
y[n] = T{x[n]} = T{

$$y[n] = T\{x[n]\} = T\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\}$$

$$= \sum_{k=-\infty}^{\infty} x[k]T\{\delta[n-k]\}$$



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$$= \sum_{k=-\infty}^{\infty} x[k]h[n,k]$$

Where h(n,k) is the response of unit impulses δ[n-k] $-\infty < k < \infty$.
If the response of the LTI system to the unit sample sequence δ[n] is denoted by h[n], i.e.
h[n] ≡ T{δ[n]}
then by the time-invariant property, the response of the system to the delayed unit sample sequence δ[n-k] is,
h[n-k] = T{δ[n-k]}
then,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

The formula in above equation that gives the response y[n] of the LTI system as a function of the input signal x[n] and the unit impulse h[n] is called a convolution sum.

**Methods to calculate Convolution Sum:**

1. Mathematically (by using direct formula of convolution sum).
2. Graphically (by using following procedure)
3. Matrix Method

**Procedure to compute the convolution sum:**
Suppose that we wish to compute the output of the system at some time instant say $n = n_0$, then

$$y[n_0] = \sum_{k=-\infty}^{\infty} x[k]h[n_0-k]$$

**The procedures are:**
Plot x[n] and h[n] as x[k] and h[k] then,

1) Folding: Fold h[k] about k = 0 to obtain h[-k].
2) Shifting: Shift h[-k] by $n_0$ to the right (left) if $n_0$ is positive (negative), to obtain $h[n_0-k]$.
3) Multiplication: Multiply x[k] by $h[n_0-k]$ to obtain the product sequence $v_{n0}(k) \equiv x[k]h[n_0-k]$.
4) Summation: Sum all the values of the product sequence $v_{n0}(k)$ to obtain the value of the output at time $n = n_0$.
5) Repetition: Repeat steps 2 through 4, for all possible time shifts $-\infty < n < \infty$ to obtain overall response.

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**Properties of Convolution:**
To simplify the notation, we denote the convolution operations as,

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

1) Commutative Law: $x[n] * h[n] = h[n] * x[n]$
2) Associative Law: $(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$
3) Distributive law: $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$

## Causal Linear Time Invariant Systems:

Causal System: whose output at time n depends only on present and past inputs but does not depends on future inputs. i.e. Output of system x[n] at instant $n = n_0$ depends on values of x[n] for $n \leq n_0$.

**Causal LTI System:**



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Let us consider an LTI system having an output at time n = $n_0$ by

$$y[n_0] = \sum_{k=-\infty}^{\infty} h[k]x[n_0 - k]$$

Suppose that we subdivide the sum into two sets of term, one involving negative terms and other involving positive terms of values of k.

$$y[n_0] = \sum_{k=0}^{\infty} h[k]x[n_0 - k] + \sum_{k=-\infty}^{-1} h[k]x[n_0 - k]$$

$=\{h[0]x[n_0] + h[1]x[n_0 - 1]+ h[2]x[n_0 - 2] + - - - - - - \} + \{h[-1]x[n_0 + 1]+ h[-2]x[n_0 +2] + - - - \}$

We observe that the terms in the first sum involve $x[n_0]$, $x[n_0 - 1]$, - - - which are the present and past values. On the other hand, the terms in the second sum involves the input signals $x[n_0 + 1]$, $x[n_0 + 2]$, - - -. Now, if the output at time n = $n_0$ is to depend only on the present and past values, then it is clear that

$h[n] = 0$ for $n < 0$

It is both a necessary and a sufficient condition for causality. Hence "*An LTI system is causal if and only if its impulse response is zero for negative values of n.*"
Thus for causal LTI system,

$$y[n] = \sum_{k=0}^{\infty} h[k]x[n - k]$$

$$y[n] = \sum_{k=-\infty}^{n} x[k]h[n - k]$$

## Stability of Linear Time Invariant Systems:

Stable System: An arbitrary relaxed system is BIBO stable if and only if its output sequence y[n] is bounded for every bounded input x[n].
For an LTI system:
Suppose we have an LTI system,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n - k]$$

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Taking absolute values on both sides,

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k]x[n - k] \right|$$

$$\leq \sum_{k=-\infty}^{\infty} |h[k]||x[n - k]|$$

If the input is bounded, there exists a finite number $M_x$ such that $|x[n]| \leq M_x$

$$|y[n]| \leq M_x \sum_{k=-\infty}^{\infty} |h[n]|$$

The output is bounded if the impulse response of the system satisfies the condition

$$S_h \equiv \sum_{k=-\infty}^{\infty} |h[n]| < \infty$$



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"*A linear time-invariant system is stable if its impulse response is absolutely summable*". This is the necessary and sufficient condition for stable LTI system.

## Systems with Finite-Duration and Infinite Duration Impulse Response:

**Finite Duration Impulse Response System (FIR system):**

h[n] = 0 for n < 0 and n ≥ M,

$$y[n] = \sum_{k=0}^{M} h[k]x[n-k]$$

FIR has a finite memory of length M-samples.

**Infinite Duration Impulse Response System (IIR System)**

$$y[n] = \sum_{k=0}^{\infty} h[k]x[n-k]$$

IIR has an infinite memory.

## Discrete-time system described by Difference Equations:

The convolution summation formula is given by

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

Above equation suggests a means for the realization of the system. In the case of FIR systems, such a realization involves additions, multiplications, and a finite number of memory location, which is readily implemented directly.

If the system is IIR, however, its practical implementation as given by convolution is clearly impossible, since it requires an infinite number of memory locations, multiplications, and additions.

There is a practical and computationally efficient means for implementing a family of IIR systems, within the general class of IIR systems; this family of discrete-time systems is more conveniently described by difference equations.

## Recursive & Non-Recursive Discrete-time systems:

If the response of any discrete-time systems depends only on the input signals (i.e. terms of the input signal) then the system is known as non-recursive discrete time system.

If we can express the output of the system not only in terms of the present and past values of the input, but also in terms of the past output values, then that system is known as recursive system.

Eg: the cumulative average of a signal x[n] in the interval 0 ≤ k ≤ n defined as,

$$y[n] = \frac{1}{n+1}\sum_{k=0}^{n} x[k] \qquad n = 0, 1, 2, \ldots..$$

The realization of above equation requires the storage of all the input samples. Since n is increasing, our memory requirements grow linearly with time.

However, y[n] can be computed more efficiently by utilizing the previous output value y[n-1]. By a simple algebraic rearrangement, we obtain,

$$(n+1)y[n] = \sum_{k=0}^{n-1} x[k] + x[n] = ny[n-1] + x[n]$$

$$\therefore y[n] = \frac{n}{n+1}y[n-1] + \frac{1}{n+1}x[n]$$



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Now the cumulative average y[n] can be computed recursively by multiplying the previous output value y[n-1] by n/(n+1), multiplying the present input x[n] by 1/(n+1), and adding the two products.
This is an example of recursive system. In general whose output y[n] at time n depends on any number of past output values y[n-1], y[n-2], ……, is called a recursive system.
The output of a causal and practically realizable recursive system can be expressed in general as,
y[n] = F{y[n-1], y[n-2], ……, y[n-N], x[n], x[n-1], ….., x[n], x[n-1], ….., x[n-M]}
If y[n] depends only on the present and past inputs, then
y[n] = F{ x[n], x[n-1], ….., x[n], x[n-1], ….., x[n-M]}
Such a system is called non-recursive.
In recursive system, we need to compute all the previous values y[0], y[1], y[2], …., $y[n_0 - 1]$ to compute $y[n_0]$ but in non-recursive system, we can compute the $y[n_0]$ immediately without having $y[n_0-1]$, $y[n_0-2]$, ……. This feature is desirable in some practical applications.

**Difference Equations:**

An LTI discrete-system can also be described by a linear coefficient difference equation of the form,

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \; a_0 \equiv 1$$

Or, equivalently,

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

If $a_N \neq 0$, then the difference equation is of order N. This equation describes a recursive approach for computing the current output, given the input values & previously computed output-values.
If the system described by difference equation has a constant coefficient (independent of time) then it is known as linear constant coefficient difference (LCCD) equation.
Consider the first order system (i.e. N = 1) & M = 0.
y[n] = ay[n-1] + x[n]
Now,
y[0] = ay[-1] + x[0]
$y[1] = ay[0] + x[1] = a^2y[-1] + ax[0] + x[1]$
$y[2] = ay[1] + x[2] = a^3y[-1] + a^2x[0] + ax[1] + x[2]$
⋮ ⋮ ⋮ ⋮
$y[n] = ay[n-1] + x[n] = a^{n+1}y[-1] + a^nx[0] + a^{n-1}x[1] + \ldots.$

$$= a^{n+1}y[-1]\sum_{k=0}^{n} a^k x[n-k] \qquad n \geq 0$$

the response contain two parts; the first part is the result of the initial condition y[-1] of the system and second part is the response of the system to the input signal x[n].
if the system is initially relaxed at time n = 0, then its memory should be zero. Hence y[-1] = 0 (state = output of the delay element).
In this case, we say that the system is at zero state and its corresponding output is called zero-state response or forced response and is denoted by $y_{zs}$.

$$y_{zs}[n] = \sum_{k=0}^{n} a^k x[n-k] \qquad n \geq 0$$



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We note that above equation is a convolution summation involving the input signal convolved with the impulse response $h[n] = a^n u[n]$.
We obtained the result that the relaxed recursive system described by the first order difference equation is a linear time-invariant IIR system with impulse response given by $h[n] = a^n u[n]$.
Now, suppose the system is initially non-relaxed (i.e. $y[-1] \neq 0$) and the input $x[n] = 0$ for all n. the output of the system with zero input is called the zero-input response or natural response and is denoted by $y_{zi}[n]$.

$$y_{zi}[n] = a^{n+1}y[-1] \quad n \geq 0$$

We observe that a recursive system with nonzero initial condition is non-relaxed in the sense that it can produce an output without being excited.
Linearity, time invariance and Stability of the system described by LCCD equation
A system is linear if it satisfies the following three requirements:

1. The total response is equal to the sum of the zero-input and zero-state responses. (i.e. $y[n] = y_{zi}[n] + y_{zs}[n]$)
2. The principle of superposition applies to the zero-state response.
3. The principle of superposition applies to the zero-input response.

Else the system is non-linear.
In general, recursive systems described by the constant-coefficient difference equation is linear and time-invariant. (because the coefficients $a_k$ and $b_k$ are constants)

## Some Trigonometric Identities:

$$\sin(2\pi + \theta) = sin\theta;\ \sin(2\pi - \theta) = -sin\theta$$

$$\sin(\pi \pm \theta) = \mp sin\theta$$

$$\cos(\pi \pm \theta) = -cos\theta$$

$$cos(2\pi \pm \theta) = cos\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$

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$$\sin(A \pm B) = sinAcosB \pm cosAsinB$$

$$\cos(A \pm B) = cosAcosB \mp sinAsinB$$

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